

Elon Musk's Neuralink aims to merge the human brain with AI

Cyborgs are no longer science fiction. The field of brain-machine interfaces (BMI)—which uses electrodes, often implanted into the brain, to translate neuronal information into commands capable of controlling external systems such as a computer or robotic arm—have been around for some time. Entrepreneur Elon Musk's company, Neuralink, is aiming to test its BMI systems on a human patient by the end of 2020.

In the long term, BMI devices may help monitor and treat symptoms of neurological disorders and control artificial limbs. But these could also provide a blueprint to design artificial intelligence (AI) and even enable direct brain-to-brain communication. However, for the time being, the main challenge

is to develop BMIs that avoid damaging brain tissue and cells during implantation and operation.

To avoid damage and immune responses, researchers are increasingly focusing on the development of so-called flexible BMI. These are much softer than silicone implants and are like actual brain tissue, according to *The Conversation*.

For example, Neuralink made its first designed flexible threads and inserter—tiny, thread-like probes, which are much more flexible than previous implants—to link a human brain directly to a computer. These were designed to minimise the chance of the brain's immune response rejecting the electrodes after insertion during brain surgery.

Meet Olli, the 3D-printed autonomous EV shuttle service



Olli, the 3D-printed EV shuttle service (Credit: <https://localmotors.com>)

Local Motors, an American automotive OEM, has introduced the 3D-printed, electric, autonomous shuttle known as Olli. Olli is ready to change the way we view transportation. It has been designed for neighbourhoods, campuses and urban city centres as a smart, safe, sustainable transportation option.

Enabled by additive manufacturing, more specifically, a big area additive manufacturing (BAAM) 3D printer from Cincinnati Inc. BAAM printer, and an IBM Watson Internet of Things (IoT) platform, this vehicle can be 3D-printed and assembled in under ten hours.

Moreover, Olli is equipped with 360-degree LiDAR scanners that can sense the surroundings in all directions. As a result of 3D printing, the vehicle can be customised for different environments and needs. Numerous road tests, including Olli on the streets of Berlin in a collaboration with Deutsche Bahn, has led to the 3D-printed shuttle's recent deployment at Sacramento State University campus.

PrinCube is the world's smallest mobile colour printer

PrinCube is the ultimate handheld, portable printer. It is ultra-lightweight, fits in the palm of your hand and lets you create fast, vivid, colour printing anywhere. It works over Wi-Fi with your phone to easily upload any text, image or design, and instantly print onto virtually any object or surface at the touch of a button.

This lightweight, colour printer is portable enough to take the power of inkjet printing on-the-go. Its colour inkjet personalises ordinary objects, makes messages that stand out, and is the perfect tool for arts and crafts. It is ideal for business, travel, events, art and fun, among others.

PrinCube works on any texture or shape, even on the skin, and can print over uneven surfaces with ease. It works with both iPhone and Android smartphones, and getting started is easy.

PrinCube works seamlessly with the smartphones and all printing operations can be controlled with a single print button. It can connect via a mobile hotspot or Wi-Fi to smartphone or computer. And you can simply select an image or text, hit the button and print away!

PrinCube is wireless, works over Wi-Fi, and is lightweight and ultra-portable
(Credit: www.indiegogo.com)

